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Experiment 4

Experiment 4: Statistical Hypothesis Testing Using Python

Aim: To perform statistical hypothesis testing on the Titanic dataset using Python and interpret the results using p-values.

Mapped CO(s): CO2

1. Import Required Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)
ALPHA = 0.05
```

2. Load the Titanic Dataset

In [2]:
```
df = sns.load_dataset("titanic")

print("Dataset loaded successfully")
print("Shape:", df.shape)
df.head()
```

```
Dataset loaded successfully
Shape: (891, 15)
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who
0	0	3	male	22.0	1	0	7.2500	S	Third	man
1	1	1	female	38.0	1	0	71.2833	C	First	woman
2	1	3	female	26.0	0	0	7.9250	S	Third	woman
3	1	1	female	35.0	1	0	53.1000	S	First	woman
4	0	3	male	35.0	0	0	8.0500	S	Third	man

3. Select Variables and Handle Missing Values

In [3]:
```
data = df[["survived", "age"]].dropna()

print("Rows after removing missing Age values:", len(data))
print("Missing values remaining:", data.isna().sum().sum())
data.head()
```

```
Rows after removing missing Age values: 714
Missing values remaining: 0
```

Out[3]:

	survived	age
0	0	22.0
1	1	38.0
2	1	26.0
3	1	35.0
4	0	35.0

4. Formulate Hypotheses

Question: Does the average age of passengers differ significantly based on survival status?

- **Null Hypothesis (H0):** There is no significant difference in mean age between survived and non-survived passengers.
  - H0: mu_survived = mu_not_survived
- **Alternative Hypothesis (H1):** There is a significant difference in mean age between survived and non-survived passengers.
  - H1: mu_survived != mu_not_survived
- **Significance level:** alpha = 0.05
- **Test:** Independent (two-sample) t-test

5. Separate Survived and Not Survived Groups

In [4]:
```
age_survived = data.loc[data["survived"] == 1, "age"]
age_not_survived = data.loc[data["survived"] == 0, "age"]

summary = pd.DataFrame({
    "Group": ["Survived", "Not Survived"],
    "Count": [len(age_survived), len(age_not_survived)],
    "Mean Age": [age_survived.mean(), age_not_survived.mean()],
    "Std Dev": [age_survived.std(), age_not_survived.std()]
})

print("Group summary:")
summary.round(3)
```

```
Group summary:
```

Out[4]:

	Group	Count	Mean Age	Std Dev
0	Survived	290	28.344	14.951
1	Not Survived	424	30.626	14.172

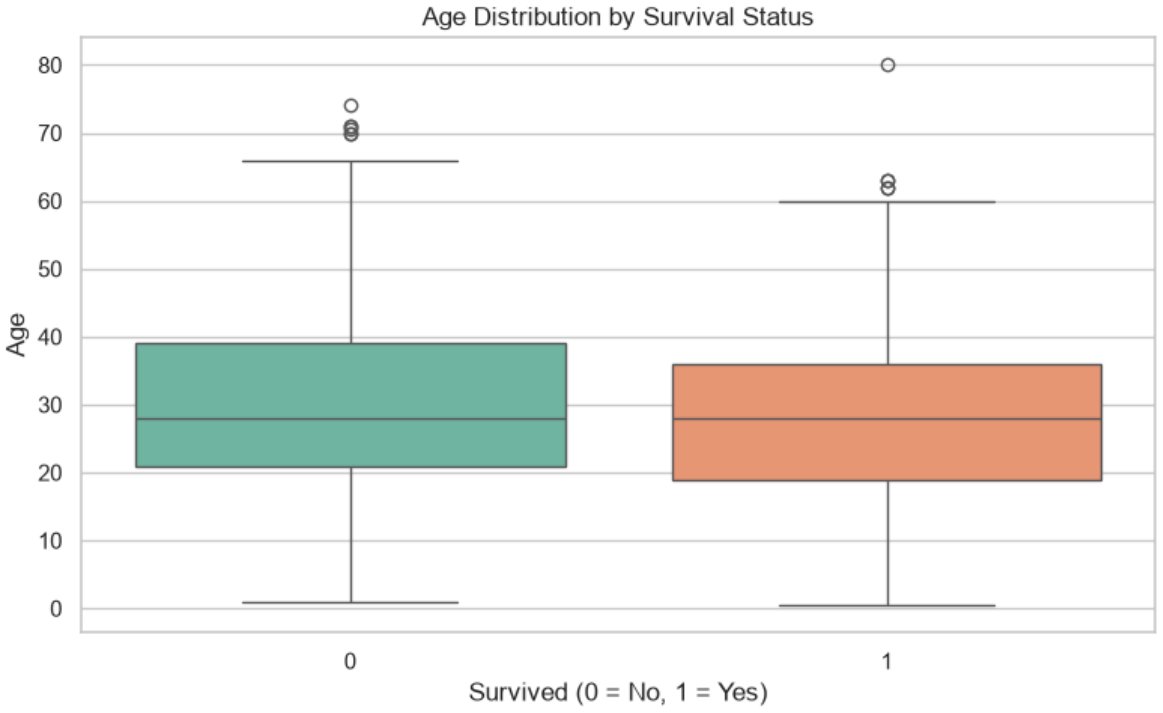
In [5]:
```
sns.boxplot(data=data, x="survived", y="age", palette="Set2")
plt.title("Age Distribution by Survival Status")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Age")
```

```
plt.tight_layout()
plt.show()
```

```
/var/folders/9l/rs7bfxps4jd0jsfdd_hrzvjr0000gn/T/ipykernel_8291/331286948
6.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for
the same effect.

  sns.boxplot(data=data, x="survived", y="age", palette="Set2")
```



6. Independent Sample t-test (Age)

In [6]:
```
t_stat, p_value = stats.ttest_ind(age_survived, age_not_survived, equal_v

print("Independent Sample t-test (Age ~ Survived)")
print(f"t-statistic = {t_stat:.4f}")
print(f"p-value     = {p_value:.6f}")
print(f"alpha       = {ALPHA}")

if p_value < ALPHA:
    print("\nDecision: Reject H0")
    print("Conclusion: There is a statistically significant difference in
else:
    print("\nDecision: Fail to reject H0")
    print("Conclusion: There is no statistically significant difference i
```

```
Independent Sample t-test (Age ~ Survived)
t-statistic = -2.0460
p-value     = 0.041190
alpha       = 0.05

Decision: Reject H0
Conclusion: There is a statistically significant difference in mean age be
tween survived and non-survived passengers.
```

Exercises / Assignments

Exercise 1: t-test using Fare instead of Age

In [7]:
```
fare_data = df[["survived", "fare"]].dropna()
fare_survived = fare_data.loc[fare_data["survived"] == 1, "fare"]
fare_not_survived = fare_data.loc[fare_data["survived"] == 0, "fare"]

print("Mean Fare - Survived:", round(fare_survived.mean(), 2))
print("Mean Fare - Not Survived:", round(fare_not_survived.mean(), 2))

t_fare, p_fare = stats.ttest_ind(fare_survived, fare_not_survived, equal_
print(f"\nt-statistic = {t_fare:.4f}")
print(f"p-value     = {p_fare:.6e}")

if p_fare < ALPHA:
    print("Decision: Reject H0 - mean Fare differs significantly by survi
else:
    print("Decision: Fail to reject H0 - no significant difference in mea
```

```
Mean Fare - Survived: 48.4
Mean Fare - Not Survived: 22.12

t-statistic = 6.8391
p-value     = 2.699332e-11
Decision: Reject H0 - mean Fare differs significantly by survival.
```

Exercise 2: Change significance level from 0.05 to 0.01

In [8]:
```
alphas = [0.05, 0.01]

print("Comparing Age t-test decisions at different alpha levels:\n")
for a in alphas:
    decision = "Reject H0" if p_value < a else "Fail to reject H0"
    print(f"alpha = {a:.2f} | p = {p_value:.6f} | {decision}")
```

```
Comparing Age t-test decisions at different alpha levels:

alpha = 0.05 | p = 0.041190 | Reject H0
alpha = 0.01 | p = 0.041190 | Fail to reject H0
```

Exercise 3: Interpret the obtained p-value

The p-value is the probability of observing a difference in sample means at least as extreme as the one obtained, assuming H0 is true.

- If **p < alpha**: evidence against H0 is strong enough -> reject H0.
- If **p >= alpha**: evidence is not strong enough -> fail to reject H0.

For the Age t-test, compare the printed p-value with 0.05 (and 0.01 in Exercise 2).

Exercise 4: Chi-Square test (Sex vs Survived)

In [9]:
```
contingency = pd.crosstab(df["sex"], df["survived"])
print("Contingency Table: Sex vs Survived")
print(contingency)

chi2, p_chi, dof, expected = stats.chi2_contingency(contingency)

print("\nChi-Square Test of Independence")
print("H0: Sex and Survived are independent")
print("H1: Sex and Survived are associated")
print(f"chi2 = {chi2:.4f}")
print(f"dof  = {dof}")
print(f"p-value = {p_chi:.6e}")

if p_chi < ALPHA:
    print("Decision: Reject H0 - Sex and survival are significantly assoc
else:
    print("Decision: Fail to reject H0 - no significant association.")
```

```
Contingency Table: Sex vs Survived
survived    0    1
sex
female     81  233
male      468  109

Chi-Square Test of Independence
H0: Sex and Survived are independent
H1: Sex and Survived are associated
chi2 = 260.7170
dof  = 1
p-value = 1.197357e-58
Decision: Reject H0 - Sex and survival are significantly associated.
```

Exercise 5: Hypotheses for a different variable (Fare)

Variable: Fare by Survival

- **H0:** Mean fare of survived passengers = mean fare of non-survived passengers.
- **H1:** Mean fare of survived passengers != mean fare of non-survived passengers.

(Already tested in Exercise 1.)

Observation

- The Titanic dataset was successfully loaded.
- Missing values in the Age column were removed.
- Passengers were divided into survived and non-survived groups.
- An Independent Sample t-test was successfully performed.
- The p-value indicated whether the difference in average age between the two groups was statistically significant.

Result

Statistical hypothesis testing was successfully performed using Python. The hypothesis was tested using an Independent Sample t-test, and the conclusion was drawn based on the calculated p-value.

Conclusion

The experiment demonstrated the application of statistical hypothesis testing using Python. The Independent Sample t-test was used to compare the mean age of survived and non-survived passengers. Based on the p-value, an appropriate statistical conclusion was made regarding the null hypothesis. Hypothesis testing is an important tool for making data-driven decisions and validating assumptions in data science.